

Distinct Values Splits

Time limit: 1.00 s

Memory limit: 512 MB

You are given an array of n integers. Your task is to count the number of ways to split the array into continuous segments such that all segments consists of distinct values.

Input

The first line has an integers n : the size of the array.

The next line has n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

Print one integer: the answer to the problem modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$

Example

Input	Output
4 1 2 1 3	6

Explanation: There are six valid splits:

- $[1], [2], [1], [3]$
- $[1], [2], [1, 3]$
- $[1], [2, 1], [3]$
- $[1], [2, 1, 3]$
- $[1, 2], [1], [3]$
- $[1, 2], [1, 3]$